



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE– (24AD3107)

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TITLE OF THE PROJECT		
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Abstract

The **E-Commerce Sales Analytics Dashboard** is a Data Science project developed to analyze online shopping sales data and provide useful information about the performance of an e-commerce business. Nowadays, online shopping websites generate a large amount of data from customer orders, products, sales, discounts, profits, and customer locations. It is difficult to understand all this information by looking at raw data. Therefore, this project aims to convert the raw sales data into simple and meaningful information using Data Science techniques.

The project starts by collecting an e-commerce sales dataset containing information such as **order details, product names, product categories, sales amount, quantity, discount, profit, customers, dates, and regions**. The collected data is cleaned before performing the analysis. Data cleaning includes removing duplicate records, handling missing values, correcting incorrect information, and preparing the data for analysis. This helps to improve the quality and accuracy of the results.

After cleaning the data, **Exploratory Data Analysis (EDA)** is performed to understand the sales information. The project analyzes questions such as which products are selling the most, which category has the highest sales, which region has more customers, which months have higher sales, and which products provide more profit. This analysis helps in finding important patterns and trends in the business data.

The project also includes an **interactive dashboard** that presents the analyzed information using charts, graphs, and tables. The dashboard displays important information such as **total sales, total profit, total orders, number of customers, top-selling products, sales by category, sales by region, and monthly sales trends**. Users can easily understand the business performance through the dashboard without having to manually study large amounts of data.

To make the project more useful as a Data Science project, **Machine Learning** can be used for sales prediction. Historical sales data is given to a Machine Learning model, which learns from previous sales patterns and predicts possible future sales. This can help the business estimate future demand and prepare products and resources accordingly.

The project can also include **customer segmentation**. Customers can be grouped according to their purchasing behavior, such as how frequently they purchase products and how much money they spend. For example, customers can be divided into high-value customers, regular customers, and occasional customers. This information can help businesses provide better offers and marketing strategies to different types of customers.

The project can be developed using **Python, Pandas, NumPy, and Machine Learning libraries** for data processing and analysis. Visualization and dashboard tools can be used to present the results in an easy-to-understand format. The combination of data analysis, visualization, and Machine Learning makes the system useful for understanding both past and future sales performance.

The main purpose of this project is to help an e-commerce business make **better decisions using its sales data**. The system can help identify the best-selling products, understand customer behavior, find profitable categories, compare regional sales, and understand monthly sales trends. Sales prediction can also help the business prepare for future demand.

In conclusion, the **E-Commerce Sales Analytics Dashboard** provides a simple way to understand large amounts of e-commerce data. It combines **data cleaning, data analysis, visualization, and Machine Learning** to convert raw sales data into useful information. The project demonstrates how Data Science can be used in the e-commerce industry to understand business performance and support better decision-making.

Signature of the Guide

HOD